БИОЛОГИЯ · 10 КЛАСС · УРОК 9

Эволюция, селекция и биоразнообразие

Основа: Biology Grade 10, Астана-кітап, 2019. Далее в этом PDF приложены все оригинальные страницы главы, включая рисунки, задания и таблицы. Русский раздел — учебный навигатор и проверочные задания, а не дословный перевод оригинала.

Как заниматься

Изучайте каждый подраздел на оригинальных страницах, используйте словарь терминов в учебнике, затем письменно отвечайте на пять вопросов. После всех подразделов выполните итоговую проверку из 40 вопросов. Проходной порог: 28 правильных ответов.

9.1 Генетическая изменчивость

Откройте раздел 9.1 в приложенных оригинальных страницах. Найдите определения, главные схемы и примеры. Сформулируйте своими словами, что означает «Генетическая изменчивость» и почему это важно.

1. Дайте определение понятия «Генетическая изменчивость».
2. Какую роль играет «Генетическая изменчивость» в живых организмах?
3. Какие ключевые структуры, этапы или компоненты связаны с темой «Генетическая изменчивость»?
4. Приведите биологический пример, относящийся к теме «Генетическая изменчивость».
5. Как можно объяснить или проверить одно из утверждений из раздела «Генетическая изменчивость»?

9.2 Факторы эволюции

Откройте раздел 9.2 в приложенных оригинальных страницах. Найдите определения, главные схемы и примеры. Сформулируйте своими словами, что означает «Факторы эволюции» и почему это важно.

1. Дайте определение понятия «Факторы эволюции».
2. Какую роль играет «Факторы эволюции» в живых организмах?
3. Какие ключевые структуры, этапы или компоненты связаны с темой «Факторы эволюции»?
4. Приведите биологический пример, относящийся к теме «Факторы эволюции».
5. Как можно объяснить или проверить одно из утверждений из раздела «Факторы эволюции»?

9.3 Доказательства эволюции

Откройте раздел 9.3 в приложенных оригинальных страницах. Найдите определения, главные схемы и примеры. Сформулируйте своими словами, что означает «Доказательства эволюции» и почему это важно.

1. Дайте определение понятия «Доказательства эволюции».
2. Какую роль играет «Доказательства эволюции» в живых организмах?
3. Какие ключевые структуры, этапы или компоненты связаны с темой «Доказательства эволюции»?
4. Приведите биологический пример, относящийся к теме «Доказательства эволюции».
5. Как можно объяснить или проверить одно из утверждений из раздела «Доказательства эволюции»?

9.4 Этапы формирования жизни на Земле

Откройте раздел 9.4 в приложенных оригинальных страницах. Найдите определения, главные схемы и примеры. Сформулируйте своими словами, что означает «Этапы формирования жизни на Земле» и почему это важно.

1. Дайте определение понятия «Этапы формирования жизни на Земле».
2. Какую роль играет «Этапы формирования жизни на Земле» в живых организмах?
3. Какие ключевые структуры, этапы или компоненты связаны с темой «Этапы формирования жизни на Земле»?
4. Приведите биологический пример, относящийся к теме «Этапы формирования жизни на Земле».
5. Как можно объяснить или проверить одно из утверждений из раздела «Этапы формирования жизни на Земле»?

9.5 Филогенетические деревья

Откройте раздел 9.5 в приложенных оригинальных страницах. Найдите определения, главные схемы и примеры. Сформулируйте своими словами, что означает «Филогенетические деревья» и почему это важно.

1. Дайте определение понятия «Филогенетические деревья».
2. Какую роль играет «Филогенетические деревья» в живых организмах?
3. Какие ключевые структуры, этапы или компоненты связаны с темой «Филогенетические деревья»?
4. Приведите биологический пример, относящийся к теме «Филогенетические деревья».
5. Как можно объяснить или проверить одно из утверждений из раздела «Филогенетические деревья»?

9.6 Видообразование

Откройте раздел 9.6 в приложенных оригинальных страницах. Найдите определения, главные схемы и примеры. Сформулируйте своими словами, что означает «Видообразование» и почему это важно.

1. Дайте определение понятия «Видообразование».
2. Какую роль играет «Видообразование» в живых организмах?
3. Какие ключевые структуры, этапы или компоненты связаны с темой «Видообразование»?
4. Приведите биологический пример, относящийся к теме «Видообразование».
5. Как можно объяснить или проверить одно из утверждений из раздела «Видообразование»?

9.7 Селекция

Откройте раздел 9.7 в приложенных оригинальных страницах. Найдите определения, главные схемы и примеры. Сформулируйте своими словами, что означает «Селекция» и почему это важно.

1. Дайте определение понятия «Селекция».
2. Какую роль играет «Селекция» в живых организмах?
3. Какие ключевые структуры, этапы или компоненты связаны с темой «Селекция»?
4. Приведите биологический пример, относящийся к теме «Селекция».
5. Как можно объяснить или проверить одно из утверждений из раздела «Селекция»?

9.8 Эволюция человека

Откройте раздел 9.8 в приложенных оригинальных страницах. Найдите определения, главные схемы и примеры. Сформулируйте своими словами, что означает «Эволюция человека» и почему это важно.

1. Дайте определение понятия «Эволюция человека».
2. Какую роль играет «Эволюция человека» в живых организмах?
3. Какие ключевые структуры, этапы или компоненты связаны с темой «Эволюция человека»?
4. Приведите биологический пример, относящийся к теме «Эволюция человека».
5. Как можно объяснить или проверить одно из утверждений из раздела «Эволюция человека»?

ИТОГОВАЯ ПРОВЕРКА · 40 ВОПРОСОВ

Вопросы открытого типа. Сверяйте ответы с учебником. За каждый полностью правильный ответ — 1 балл; 28/40 — зачёт.

1. Объясните основной механизм, рассматриваемый в теме «Генетическая изменчивость».
2. Объясните основной механизм, рассматриваемый в теме «Факторы эволюции».
3. Объясните основной механизм, рассматриваемый в теме «Доказательства эволюции».
4. Объясните основной механизм, рассматриваемый в теме «Этапы формирования жизни на Земле».
5. Объясните основной механизм, рассматриваемый в теме «Филогенетические деревья».
6. Объясните основной механизм, рассматриваемый в теме «Видообразование».
7. Объясните основной механизм, рассматриваемый в теме «Селекция».
8. Объясните основной механизм, рассматриваемый в теме «Эволюция человека».
9. Сравните два явления или компонента, упомянутых в теме «Генетическая изменчивость».
10. Сравните два явления или компонента, упомянутых в теме «Факторы эволюции».
11. Сравните два явления или компонента, упомянутых в теме «Доказательства эволюции».
12. Сравните два явления или компонента, упомянутых в теме «Этапы формирования жизни на Земле».
13. Сравните два явления или компонента, упомянутых в теме «Филогенетические деревья».
14. Сравните два явления или компонента, упомянутых в теме «Видообразование».
15. Сравните два явления или компонента, упомянутых в теме «Селекция».
16. Сравните два явления или компонента, упомянутых в теме «Эволюция человека».
17. Как изменение условий повлияет на процесс из темы «Генетическая изменчивость»?
18. Как изменение условий повлияет на процесс из темы «Факторы эволюции»?
19. Как изменение условий повлияет на процесс из темы «Доказательства эволюции»?
20. Как изменение условий повлияет на процесс из темы «Этапы формирования жизни на Земле»?
21. Как изменение условий повлияет на процесс из темы «Филогенетические деревья»?
22. Как изменение условий повлияет на процесс из темы «Видообразование»?
23. Как изменение условий повлияет на процесс из темы «Селекция»?
24. Как изменение условий повлияет на процесс из темы «Эволюция человека»?
25. Приведите пример практического применения знаний по теме «Генетическая изменчивость».
26. Приведите пример практического применения знаний по теме «Факторы эволюции».
27. Приведите пример практического применения знаний по теме «Доказательства эволюции».
28. Приведите пример практического применения знаний по теме «Этапы формирования жизни на Земле».
29. Приведите пример практического применения знаний по теме «Филогенетические деревья».
30. Приведите пример практического применения знаний по теме «Видообразование».
31. Приведите пример практического применения знаний по теме «Селекция».
32. Приведите пример практического применения знаний по теме «Эволюция человека».
33. Назовите возможную ошибку в понимании темы «Генетическая изменчивость» и исправьте её.
34. Назовите возможную ошибку в понимании темы «Факторы эволюции» и исправьте её.
35. Назовите возможную ошибку в понимании темы «Доказательства эволюции» и исправьте её.
36. Назовите возможную ошибку в понимании темы «Этапы формирования жизни на Земле» и исправьте её.
37. Назовите возможную ошибку в понимании темы «Филогенетические деревья» и исправьте её.
38. Назовите возможную ошибку в понимании темы «Видообразование» и исправьте её.
39. Назовите возможную ошибку в понимании темы «Селекция» и исправьте её.
40. Назовите возможную ошибку в понимании темы «Эволюция человека» и исправьте её.

ОРИГИНАЛЬНАЯ ГЛАВА С ИЛЛЮСТРАЦИЯМИ

Следующие страницы воспроизведены из загруженного пользователем учебника без изменения содержания.

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CHAPTER 9.0

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CHAPTER 9

EVOLUTION AND BREEDING. BIODIVERSITY

1. Genetic variation
2. Factors of evolution
3. Evidence of evolution
4. Stages of the formation of life on Earth
5. Phylogenetic trees
6. Speciation
7. Selective breeding
8. Human evolution



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9.1 Genetic variation

You will:

- explain the relationship between genetic variation and evolution.

Key terms

- Genetic variation - the difference or diversity in gene alleles in a population;
- Mutation - a change in DNA of an organism;
- Recombination - shuffling existing alleles to produce new combinations of genes.

STQ

Which factors are necessary for a population to evolve?

Facts

Cheetahs were almost extinct during ice-age. Modern cheetahs are descendants of a small portion of cheetahs, which survived at that time. That is why cheetahs have low genetic variations and they look similar.

Text

Mutations in different genes of an organism can cause changes in this organism. Mutations in cells of ovaries and testes may be inherited by offspring. If this cell divides to form gametes, these gametes will have a mutated gene. Then one of the gametes fuses with another gamete to form a zygote. The zygote will divide, and all the cells of an organism formed from this zygote will have this mutated gene. This mutation can cause a change in phenotype of the organism. Consequently, genes and phenotype in one population will become different. This will cause genetic variation in a population.

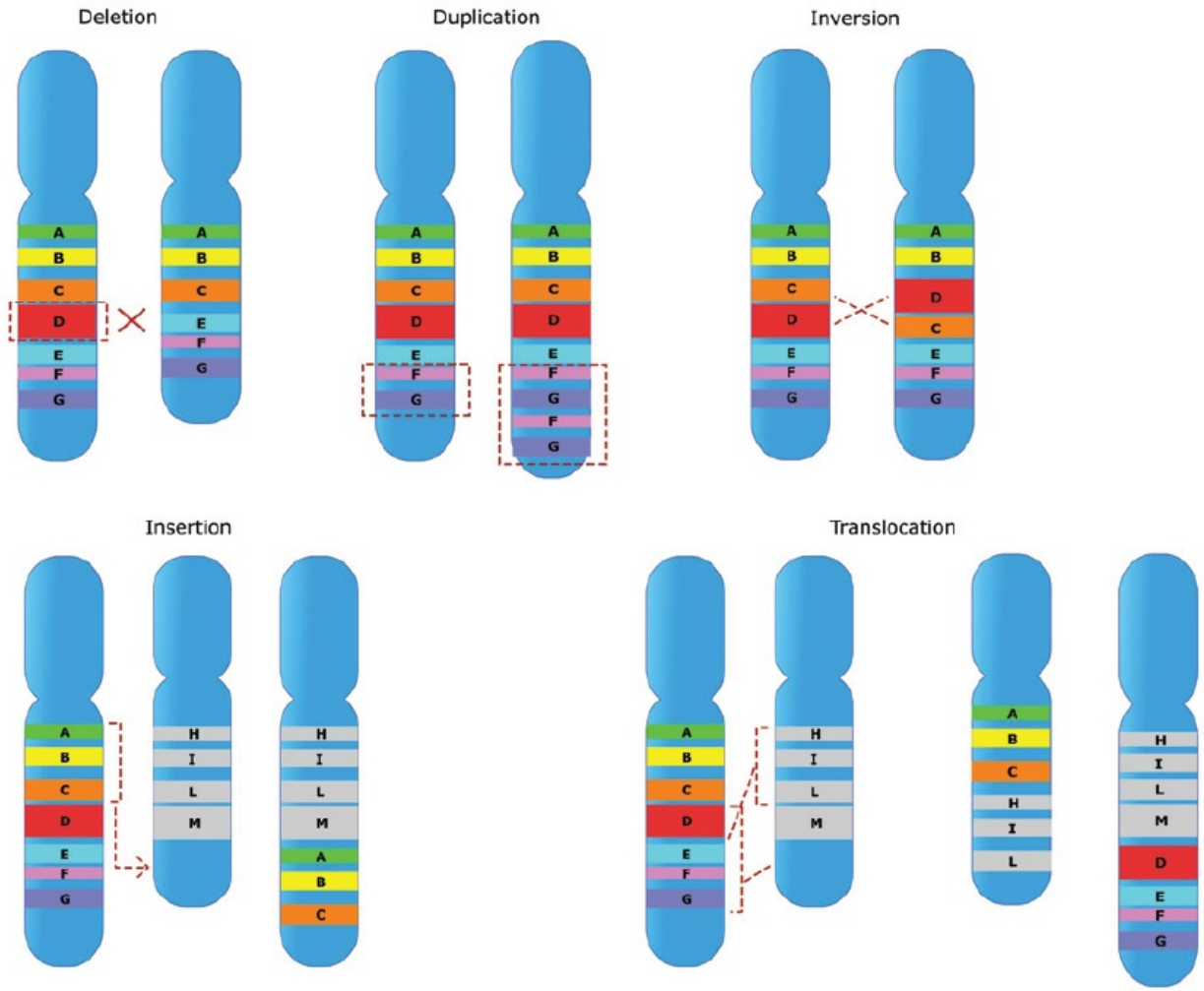
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Genetic variation is the difference or diversity in gene alleles in a population. Genetic variation gives raw material for evolutionary processes. Change in genes of an individual can be inherited by its offspring. This will change the frequency of gene alleles in all population and phenotype of the population will be different. So, this population will change and evolve. If there is no genetic variation, a population will always be the same and will not evolve.

Causes of genetic variation

Genetic variation can be caused by a change in the genotype of an organism and formation of new alleles. New alleles can be made by mutations. Mutation is a change in DNA of an organism.



Chromosome mutations

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Mutations can be beneficial or harmful. If the mutation is beneficial, organisms with new mutant allele will dominate, and frequency of this allele will increase. So, the population will change. Alternatively, if the mutation is harmful, it can be “hidden” in heterozygous organisms as a recessive allele. So, the phenotypic variation of the population will be the same, but genetic variation will be higher. When environment changes this harmful allele can become beneficial.

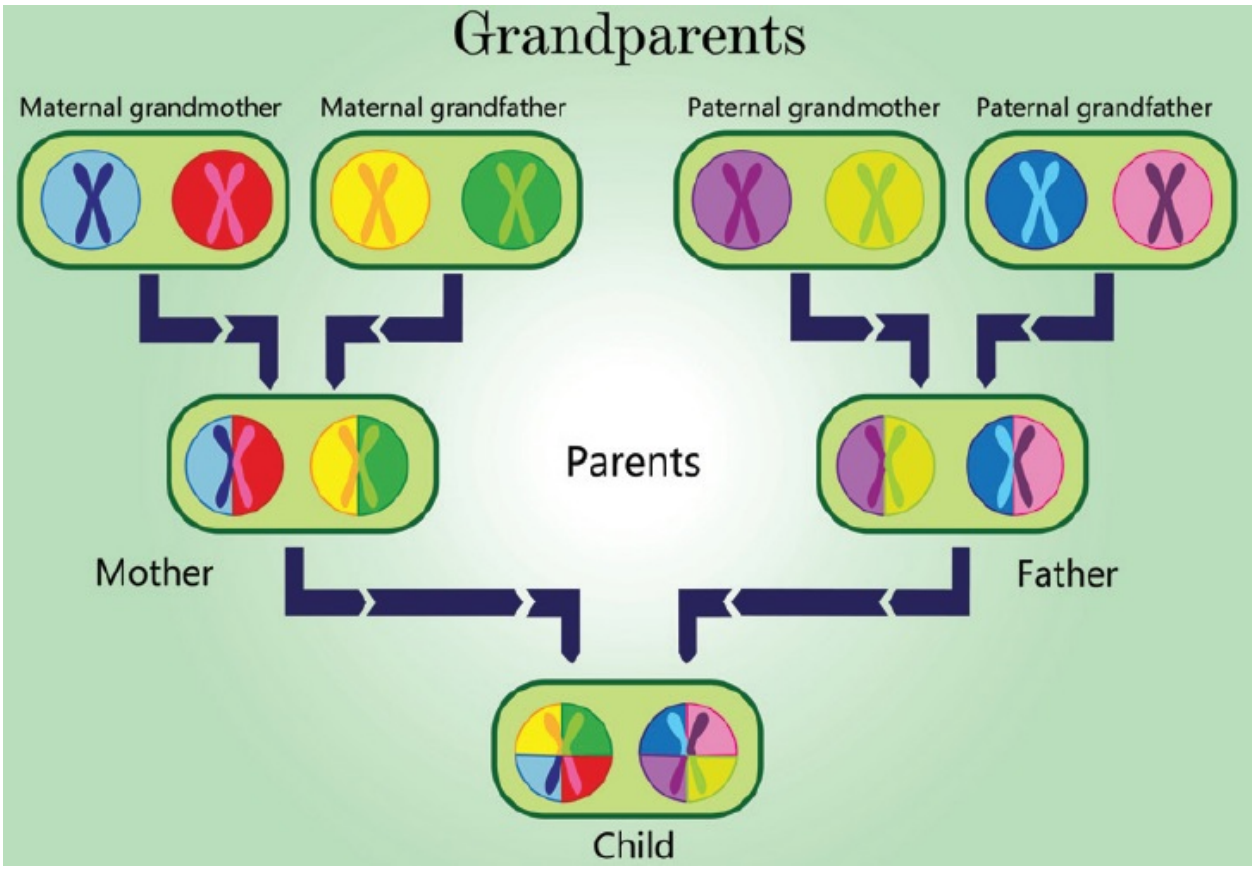
The second cause of genetic variation is recombination. Recombination is shuffling existing alleles to produce new combinations of genes during sexual reproduction. Recombinations can occur by three processes:

1. Crossing-over. Gametes are produced by meiosis. In the first step of meiosis, prophase I, two homologous chromosomes exchange parts of their genetic material. This process is called crossing-over. So, new sets of genes in different chromosomes are produced.
2. Independent assortment. Gametes have 23 chromosomes, but cells, they are produced from, have 46 chromosomes. When gametes are formed, 46 chromosomes are separated into four cells by 23. So, chromosomes are distributed randomly into these four gametes. This results in different gene combinations in one cell.
3. Fertilization. Male and female gametes fuse to produce a new organism. This process is called fertilization. During fertilization gametes with different gene sets combine in one cell and, consequently, new combinations of genes are formed.

Mutations and recombinations increase genetic variation making the evolution of different organisms and populations possible.

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Activity



Population A

Population B

Compare two given horse populations and answer the following questions:

1. Which population has higher genetic variation? Explain your answer.
2. Which population is more adapted to the conditions on the picture? Explain

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your answer.
3. Can population B be derived from population A? Explain why and how is it possible or not.

Literacy

1. Why is genetic variation necessary for evolution?
2. How can you increase genetic variation in the population?
3. Why a child never looks 100% like his father or mother?

Research time

There are two types of reproductions: asexual and sexual. What is the difference between them? Why can only sexual reproduction take place to increase genetic variation? Do a research and write a short essay.

Terminology

- beneficial - тиімді / выгодный;
- derive - пайда болу / происходить;
- distribute - таралу / распространять;
- diversity - әртүрлілік / разнообразие;
- evolve - эволюциялық даму / эволюционировать;
- exist - бар болу / существовать;
- genetic variation - тұқым қуалайтын өзгергіштік / наследственная изменчивость;
- independent assortment - тәуелсіз тұқым қуалау/ независимое наследование;
- involve - еліктіру / вовлекать;
- ovary - аналық без / яичник;
- raw - шикі / сырой;
- recombination - рекомбинация;
- shuffle - араластыру / перемешивать;
- testes - аталық ұрық безі / семенники;

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9.2 Factors of evolution

You will:

- analyze the factors that affect the evolution process.

Key terms

- Struggle for existence - competition between organisms of a population;
- Natural selection - the process, when more adapted individuals survive and reproduce;
- Genetic drift - is the change in the frequency of an existing allele in a population due to random sampling of organisms.

STQ

Why populations change over time?

Facts

Biston betularia is a moth common to UK and Ireland. Before 1849 all the moth had speckled appearance, because they were mostly living on trees covered with gray, brown and green lichens. During industrialization period air became polluted and lichens did not live on trees. So, barks of trees became darker and Biston betularia changes its appearance from speckled to black.

Text

Genetic variation is raw material of evolution. Higher the genetic variation is, higher is the chance of evolution. However, there is a second important factor necessary for evolution to occur. It is altering the frequency of different alleles. It can be affected by change in the environment, scarce of food, competition, etc. If environment changes, allele, which was beneficial before, can become less beneficial. Alternatively, the allele, which was not beneficial, can become

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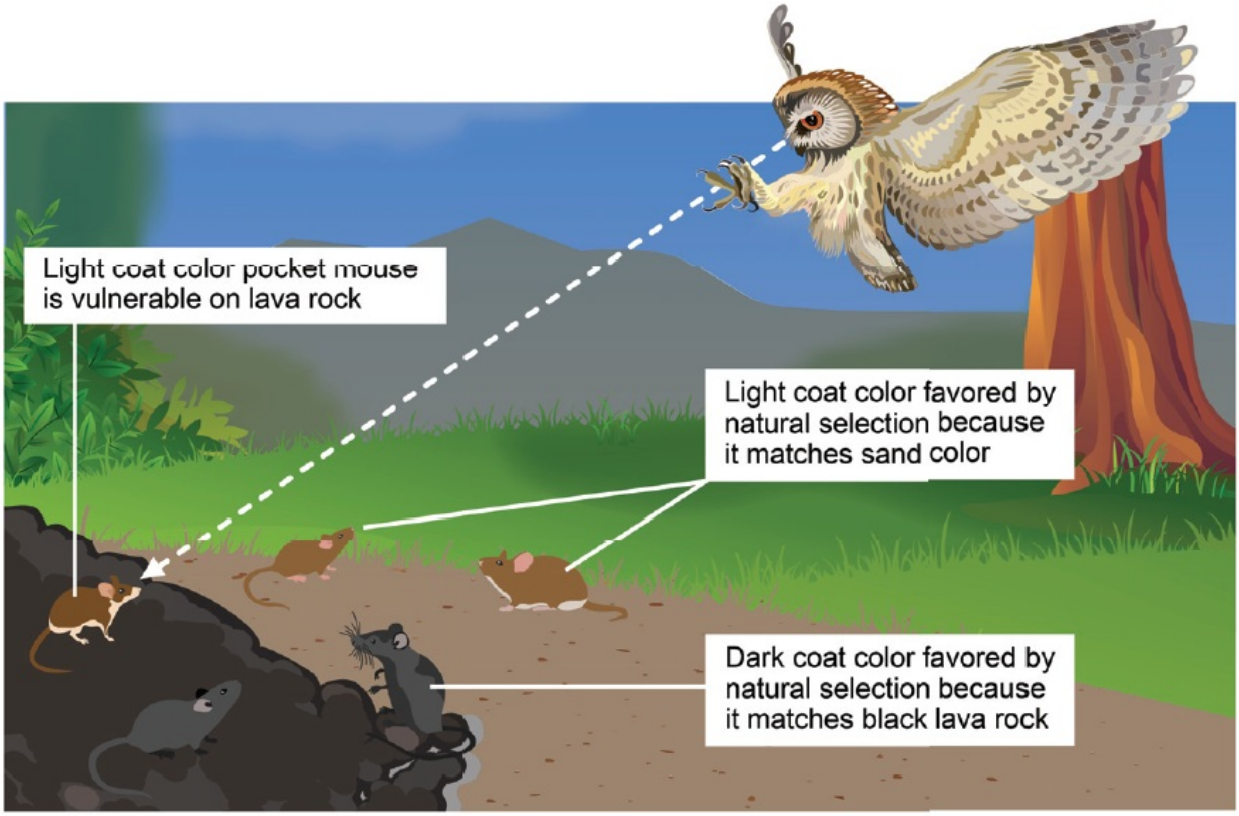
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beneficial and start dominating in the population.

Natural selection

In normal population, number of offspring produced is higher than number of adults in the population. So, some younger individuals die before reaching the adult age. It can happen due to competition for food, light, water, or by attacks of predators or parasites. Competition for area, food and other factors necessary for life is called struggle for existence.

During the struggle for existence more adapted individuals survive and reproduce. Consequently, genes of only these individuals are inherited by offspring in the population. This process is called natural selection. Due to natural selection, the proportion of alleles given to the next generations differs from the starting proportion of alleles in the population and genetic variation of the population changes.



Natural selection for gray mice over yellow mice

Genetic drift

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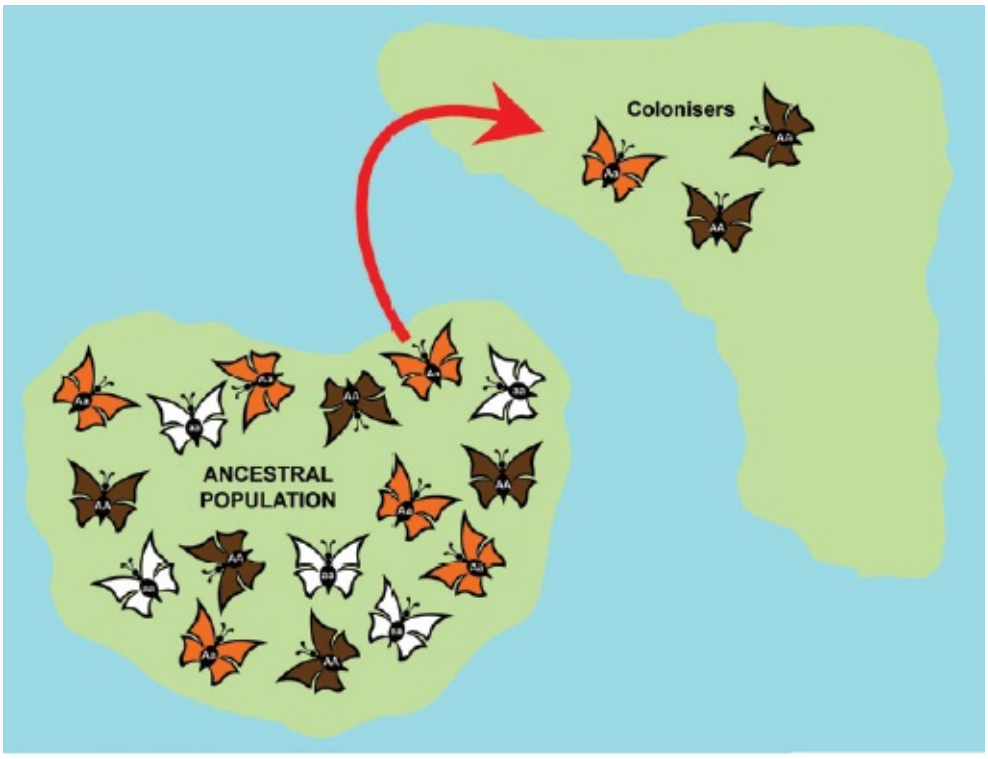
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During natural selection, genetic variation changes gradually from one generation to another. Sometimes genetic variation in the population can change sharply. For example, after a disastrous fire in the forest, most of the squirrels die. After that, some alleles can become extinct and genetic variation will decrease. This sharp change in genetic variation is called genetic drift.

During genetic drift allele frequencies can change randomly. It is impossible to predict which alleles will survive, which will become extinct. Sometimes only one allele can survive in the population and genetic variation will be low.

There are two types of genetic drift:

1. Founder effect. When a few individuals become isolated from a population, they start reproducing with each other forming a new population. Genes of this population will differ from the original population. This is called the founder effect.



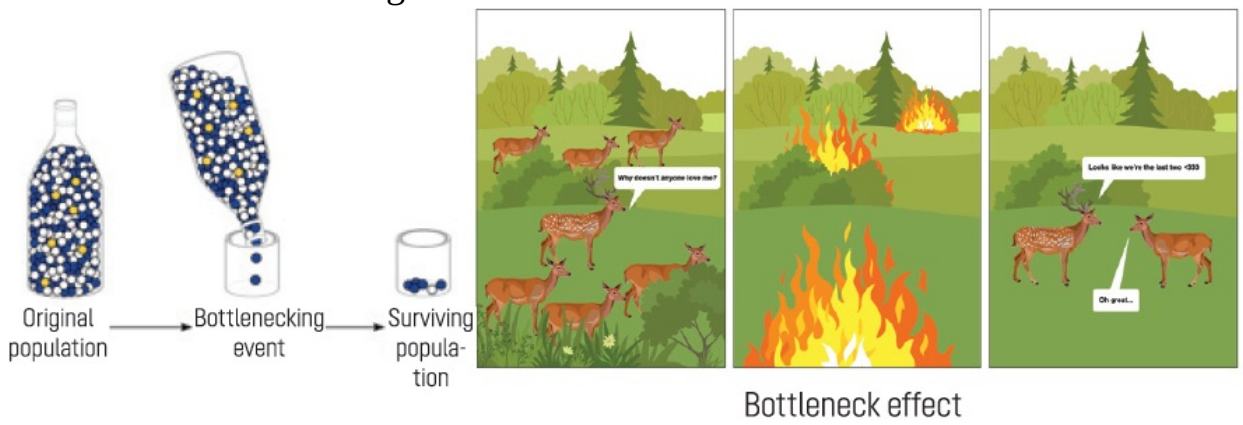
Founder effect

2. Bottleneck effect. A sudden catastrophe, such as fire or flood, can dramatically decrease the population size. So, only individuals that survived can reproduce and pass their genes to offspring. Consequently, genetic variation of

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the new population will be different from the original population and the number of individuals increases again. This is called the bottleneck effect.



Bottleneck effect

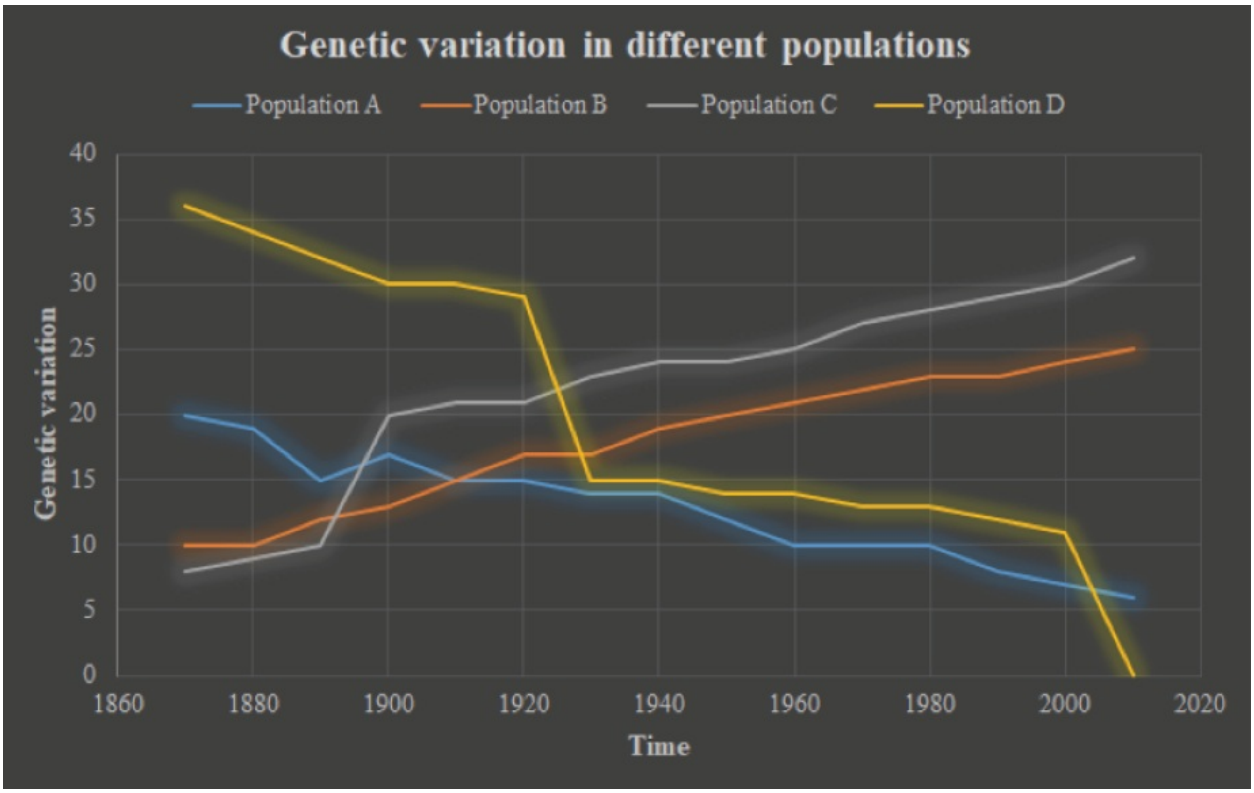
Population waves

In natural conditions population size changes. These changes are called population waves. Sometimes due to lack of food, epidemy or other factors population size can change. As a result, allel frequency in a particular population aloso can change.

Activity

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On the given graph, you can see genetic variation in four populations A, B, C, and D. Analyze the graph and explain what happens to the populations. Name the processes which affected the genetic variation in given populations.

Literacy

1. What is the difference between natural selection and genetic drift?
2. How does genetic drift affect struggle for existence?
3. Which mechanisms can increase genetic variation in a population after the genetic drift?

Research time

People used to think that the strongest species survives, but Charles Darwin refuted this statement. Why he refuted this statement and what did he state?

Terminology

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- alter - өзгерту / изменять
- bottleneck - бөтелкенің аузы / бутылочное горлышко;
- competition - бәсекелестік / конкуренция
- extinct - қырылып қалу / вымерший;
- founder - негізін қалаушы / основатель;
- genetic drift - гендердің жылжуы /дрейф генов
- gradually - біртіндеп / постепенно;
- natural selection - табиғи сұрыптау / естественный отбор
- predict - болжамдау / прогнозировать;
- refute - жоққа шығару / опровергать;
- scarce - сирек / дефицит
- sharply - кенет / резко;
- struggle for existence- тіршілік үшін күрес /борьба за существование

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9.3 Evidence of evolution

You will:

- analyze evidence of evolution.

Key terms

- Homology - similarity resulting from common ancestor.

STQ

Why whales are considered relatives of mammals and not fish?

Facts

There are structures, which have the same functions but are not derived from the common ancestor. For example, wings of a bat, dragonfly and bird. They are called analogous structures.

Text

In 1835, Charles Darwin traveled to Galapagos Islands. There he collected different types of small birds, finches. Darwin did not know much about birds, so, he thought he had collected different birds. When he went back to England, ornithologist John Gould helped him to identify, that these birds were relatives differing only in their beaks. After that, Darwin proposed that these birds had been feeding on different food, and that natural selection affected their beaks, so that new species of birds derived.

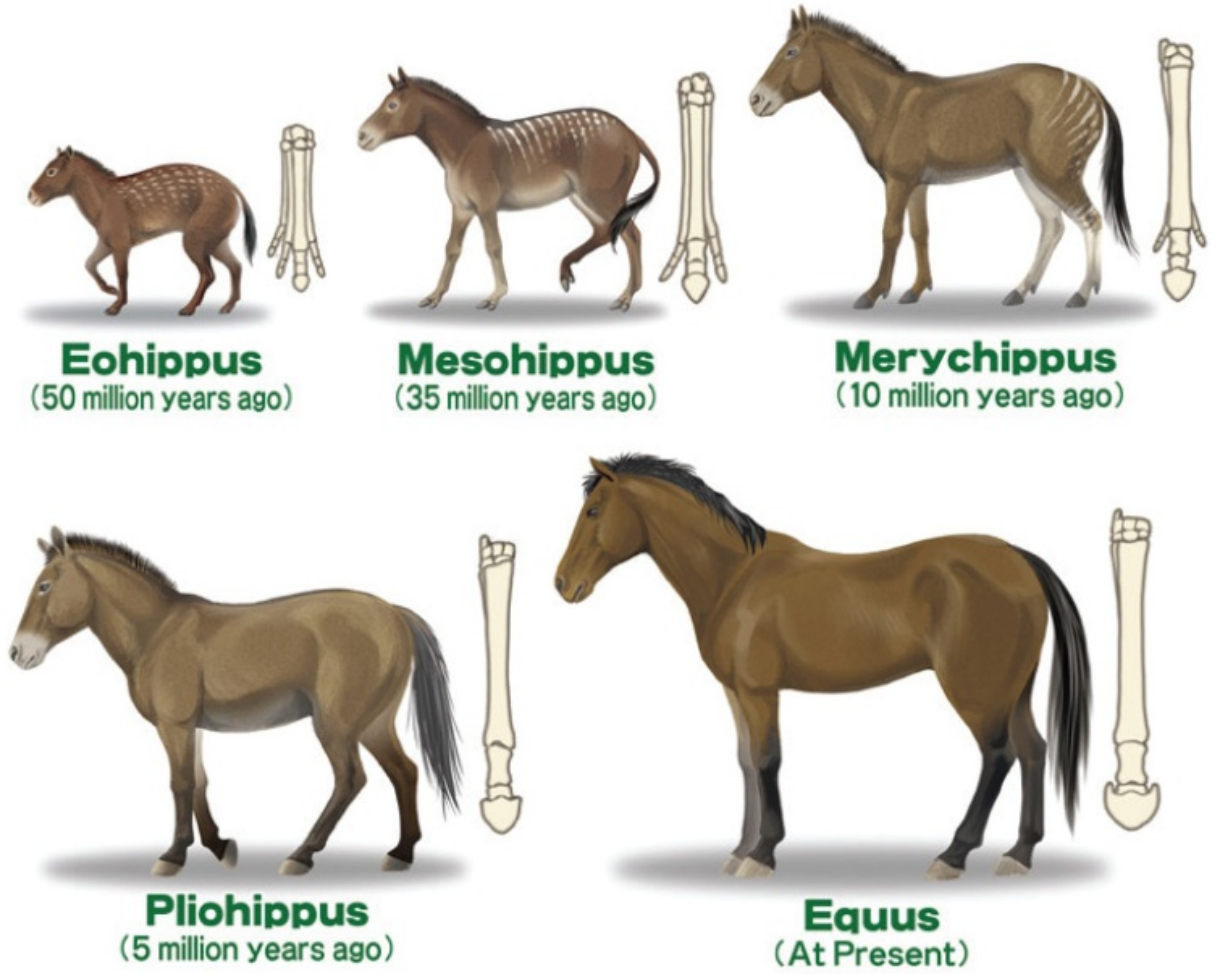
When Charles Darwin first wrote his “Origin of Species” book and proposed evolution theory based on these finches, hid did not have enough evidence. His statement was based on his observations and logic. After his theory, people started finding different evidence of evolutionary theory.

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Paleontological evidence

Paleontology is the study of fossils. Fossils give us information about species which were living many years ago. Analysis of fossil records can give us information about the development of different species through years.
For example, the study of fossil records of horses shows that ancestors of horses were smaller and had several toes. These small horses were living in the forests. Though, decreasing area of forests and increasing areas of open fields induced a change in horse structure. Horses became bigger and stronger because of running from predators in these open places.



Evolution of horses

Biogeographical evidence

Biogeography is the scientific study of the distribution of species. Knowing the

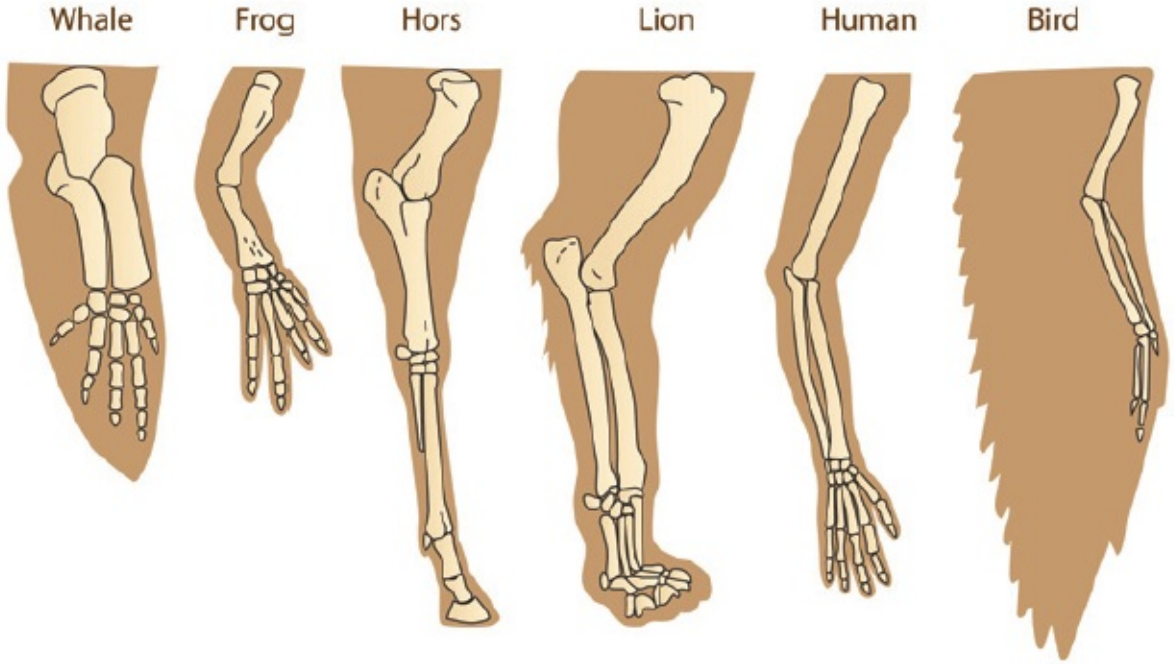
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area, where particular species lived before and live now, can show us where some species were derived from. Also, by knowing the geography scientists can explain why some species are found in only particular regions.
For example, Australia is mostly dominated by marsupial animals, whereas other parts of the world are dominated by placental animals. It is because Australia was separated from other continents 70 million years ago, and marsupials and placentals occurred and evolved separately simultaneously in Australia and other continents, and they stayed isolated by the ocean for all this time.

Homology

Characteristics of different animals can be changed by the force of natural selection. Consequently, some organs may look different and can have different functions, but they can have similar internal structure too. Similarity resulting from the common ancestor is called homology.



Forearms of different mammals

1. Anatomical homologies. Anatomical homology compares organs of different species. Homologous structures are structures that are similar because they are derived from a common ancestor. The example of homologous structures are forearms of different vertebrates. They have different appearance and functions: for walking, swimming, flying, etc, but they have the same structure of bones, which changed over time by natural

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selection adapting to their environment and functions. It shows that these forearms were derived from one common ancestor.

2. Developmental homologies. Species, which look different in adult forms, can be similar during early embryonic stages. This is called developmental homology. Early developmental stages of vertebrates have organs, which are not seen in adult organisms. For example, a human embryo has gill ridges, however, they accept oxygen through the placenta. Also, the human embryo has a bony tail, which is reduced and is not observed in adult forms.
3. Molecular homology. Similarities between genes of different species are called molecular homology. Scientists compare DNA sequences of different species. Same genes are often found in diverse organisms. The degree of similarity between different organisms shows the relatedness between organisms.

Activity

Scientists state that some organisms derived from others. Use evidence from the book to support or refute their theory about organisms given below. Which evidences you could or could not use? Explain your answers.

a. Birds from reptiles;

b. Amphibia from fish;

c. Eukaryotes from prokaryotes;

d. Whales from hippopotamus.

Literacy

1. Which evidence of evolution was found the first? Explain your answer.
2. Which evidence of evolution is the most accurate? Explain your answer.

Research time

Despite the evidence of evolution, not all scientists support the evolution theory. Do a research and find scientific objection to evolution and compare them with

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evidence.

Terminology

- accurate - дұрыс / точный;
- ancestor - ата-тек / предок;
- beak - тұмсық / клюв;
- evidence - дәлел / доказательство;
- finch- шұбар шымшық / зяблик;
- fossil - қалмақалдық / ископаемое;
- homologous - гомологты / гомологичные;
- induce - тудыру / вызвать;
- marsupial - қалталылар /сумчатые;
- objection - қарсылық / возражение.
- placental - ұрық жолдасты / плацентарный;
- toe - тұяқ / копыто;

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9.4 Stages of the formation of life on Earth

You will:

- describe the scheme and stages of the formation of life on Earth.

Key terms

- Eon - a major division of geological time, subdivided into eras;
- Era - a major division of time that is a subdivision of an eon.

STQ

How was life on Earth developing through the years?

Facts

Most of the organisms have evolved into other species, but there are organisms, like Cnidaria and Bryophyta, which did not evolve into anything.

Text

4.6 billion years ago.

Formation of Earth.

3.5 billion years ago.

Earliest forms of life, prokaryotes inhabited the Earth.

2.7 billion years ago.

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First photosynthetic prokaryotes were formed. They started producing oxygen gas (O2) in the Ocean. Then this oxygen diffused into the atmosphere and changed the Earth dramatically.

1.8 billion years ago.

First eukaryotic organisms occured. By theory of endosymbiosis, they were formed, when one bacterium has eaten another. However, the second bacteria was not digested but became useful for the first bacteria and became an organelle. In a such way, prokaryotes evolved into eukaryotes.

1.2 billion years ago.

Early small multicellular algae inhabited the Earth.

600 million years ago.

Larger multicellular organisms appeared.

535-525 million years ago.

Many present marine animal species appeared suddenly. This phenomenon is called Cambrian explosion.

500 million years ago.

The first mass extinction, or Permian extinction, happened. Volcanic eruptions caused a series of catastrophic events, that killed almost 96% of marine animals.

420 million years ago .

Colonization of land started. Plants and animals started migrating from Oceans to the land.
First reptiles and most of modern insects occurred. Vascular plants formed. Gymnosperms were dominating plant phylum. Dinosaurs evolved and first mammals appeared.
Angiosperms appeared.

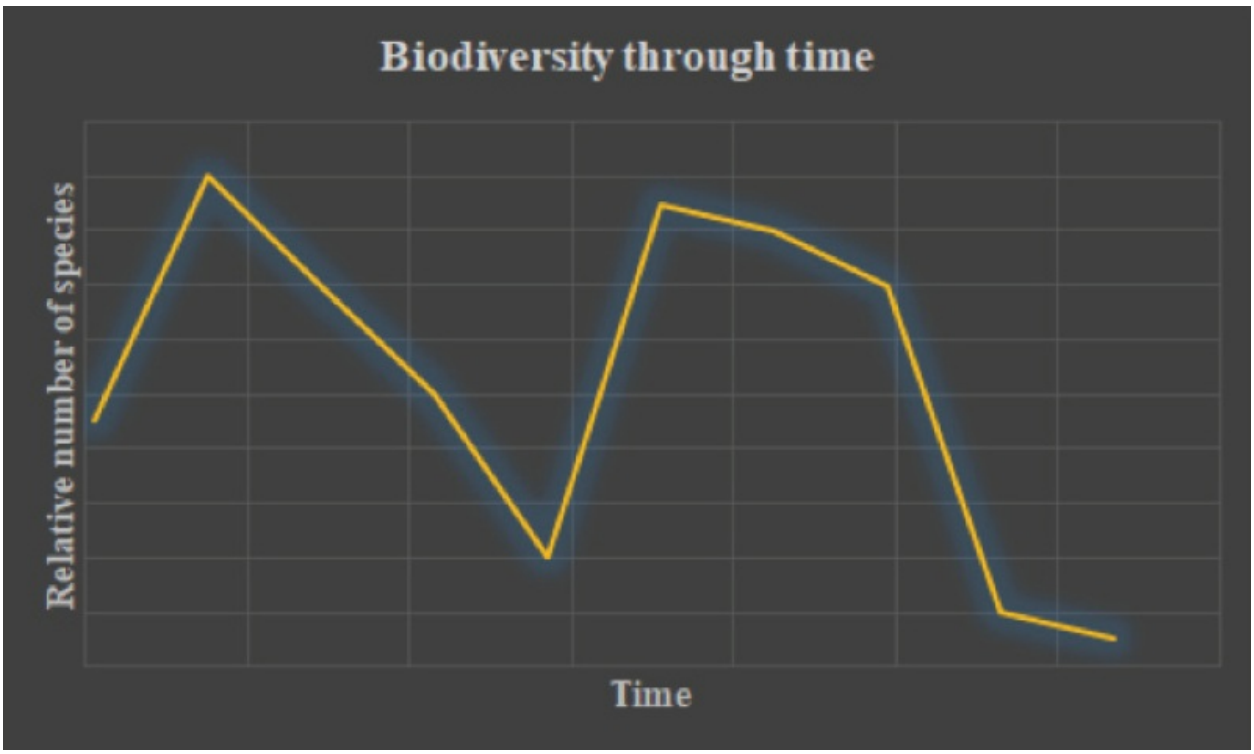
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66 million years ago.

The Cretaceous mass extinction occurred. An asteroid or large comet collided with the Earth. It produced a cloud of iridium, which eclipsed the sun. Due to this most of the animals, such as dinosaurs, and plants died.
Mammals, birds, and pollinating insects dominated the land.
Ice age, origin of genus Homo.

Activity



On the graph above, you can see changes in the number of species in different time.

1. Analyze the graph and write approximate years on the X-axis of the graph.
2. Label approximate eons and eras on the graph.
3. Explain what happened during each era.
4. Which organisms inhabited the Earth during different time on the graph?

Literacy

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1. Production of oxygen by photosynthetic organisms triggered massive extinction of prokaryotes. Why did it happen?
2. What were characteristics of organisms, which migrated to land?
3. What are advantages of living in water?

Research time

During each era different organisms were dominating the Earth. Find which organisms dominated in each era and explain what helped these organisms to dominate in each era.

Terminology

- collide - қақтығысу / сталкиваться;
- eclipse - тұтылу / затмение;
- endosymbiosis - эндосимбиоз;
- eruption - атқылау / извержение;
- explosion - жарылыс / взрыв;
- inhabit - мекен ету / обитать;
- iridium - иридий;
- trigger - түрткі болу / вызывать.

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9.5 Phylogenetic trees

You will:

- compile and interpret phylogenetic maps;
- compare the principles of different forms of phylogenetic maps.

Key terms

- Phylogenetic tree - a diagram showing the evolutionary interrelations of a group of organisms derived from a common ancestral form;
- Cladogram - a branching diagram showing the cladistic relationship between a number of species;
- Taxon - is a group of one or more populations of an organism or organisms seen by taxonomists to form a unit.

STQ

How organisms are related evolutionary?

Facts

The root of the phylogenetic tree including all organisms shows the last universal common ancestor (LUCA). It is not the first organism on the Earth, but a population of organisms all modern species originated from. There were no fossil records of the LUCA, so scientists study it by comparing genomes of nowadays organisms.

Text

Different organisms have evolved from one another. Some of them evolved from the same organism, called common ancestor. These organisms are considered relatives.

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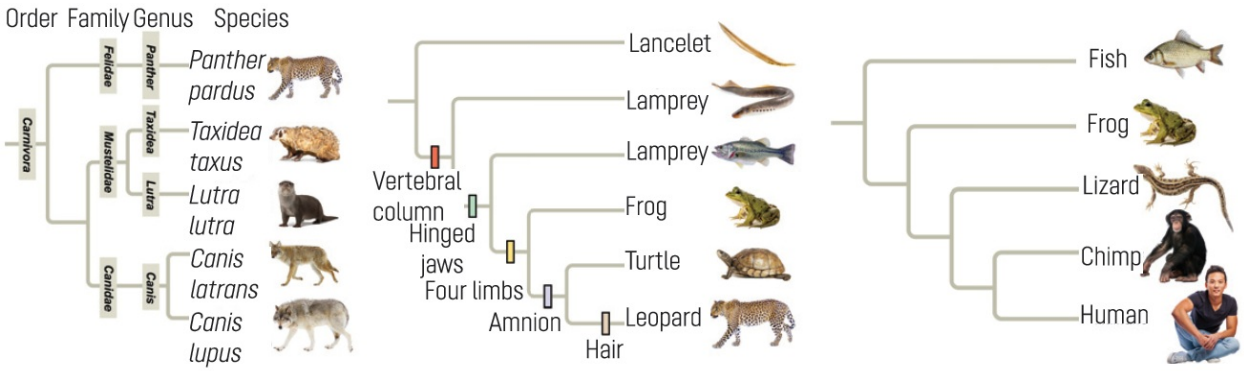
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Relatedness of organisms can be represented as a branching diagram, called a phylogenetic tree. This diagram will group organisms, which have the common ancestor, according to their systematics. The length of different branches represents the time each group of organisms evolved.

Groups of organisms on one phylogenetic tree can be put into more inclusive groups. Each branch of the tree shows a particular taxon, organisms that share the closest common ancestor. Branching points of the tree represent common ancestors.

Some phylogenetic trees are based on the evolutionary changes within the organisms and shows shared characteristics among these species. These types of phylogenetic trees are called cladograms. In cladograms length of branches is always the same and does not represent anything.

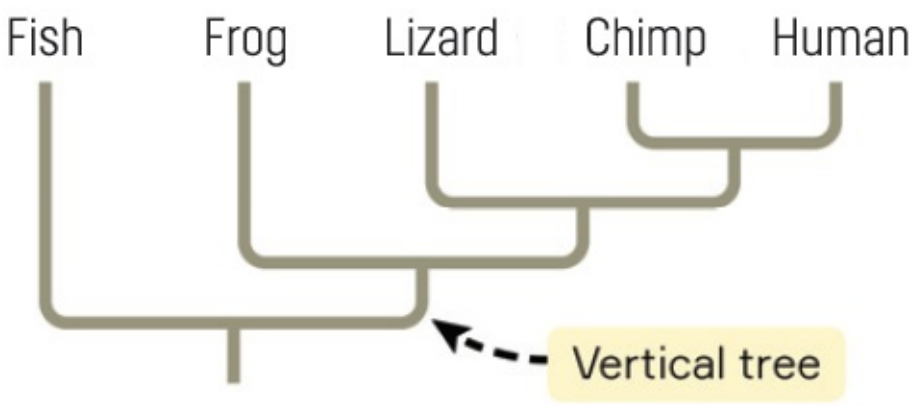
There are different forms of phylogenetic trees. They can be diagonal, vertical, or horizontal.



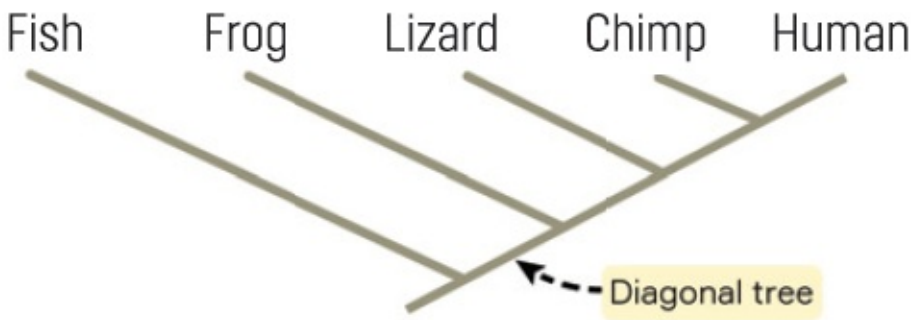
Phylogenetic tree | Cladogram | Horizontal phylogenetic tree

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Vertical phylogenetic tree



Diagonal phylogenetic tree

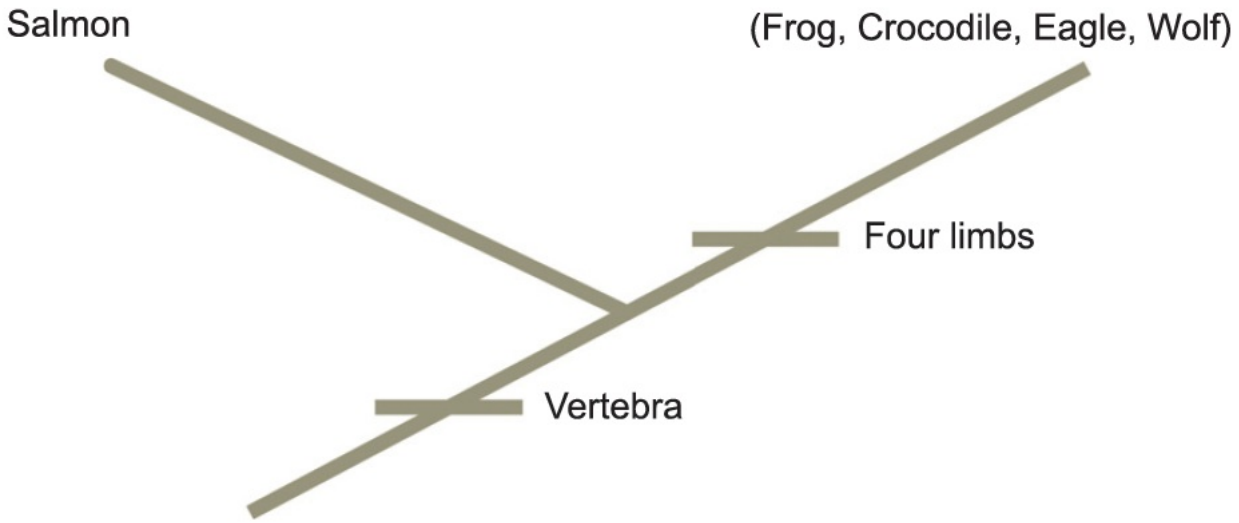
To construct a phylogenetic tree, you need to compare the structures or genomes of the organisms. Look at the table below.

Trait	*Salmon*	*Frog*	*Crocodile*	*Eagle*	*Wolf*
Vertebra	+	+	+	+	+
Four limbs	-	+	+	+	+
Amniotic egg	-	-	+	+	+
Warm-blooded	-	-	-	+	+
Hair	-	-	-	-	+

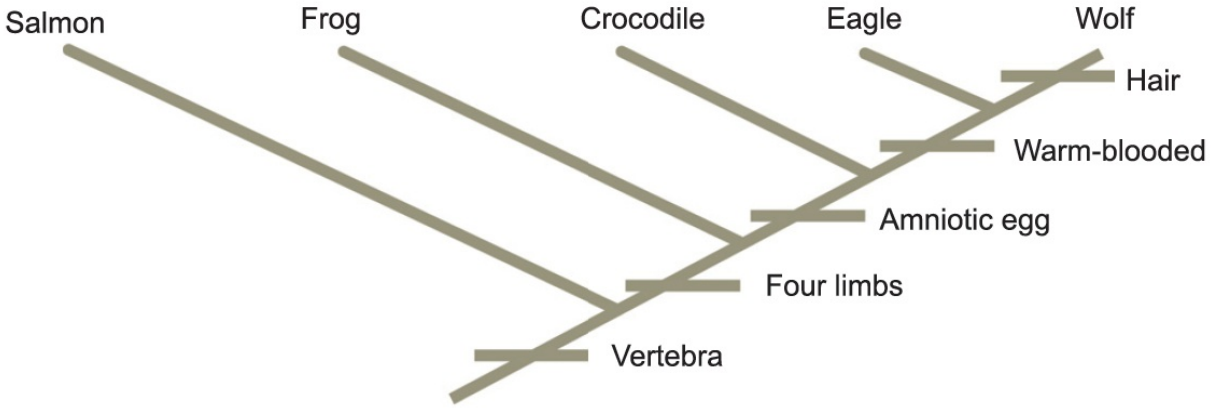
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First of all, we need to find a trait common to all the organisms. Looking at the table, we can see that vertebra is shared by all organisms. So, we can assume that vertebra is derived from the common ancestor of these organisms.
After that, we should find the trait shared by most of the organisms. In our example, frog, crocodile, eagle and wolf have four limbs, but salmon does not. We can branch off the salmon from other organisms.



Following the same pattern, we can now look for the derived trait shared by the next-largest number of organisms. This will be amniotic egg shared by crocodile, eagle and wolf. We can continue looking at derived traits and finish the tree.



Activity

Construct a phylogenetic tree using the table below.

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Trait	*Algae*	*Mosses*	*Fern*	*Pine*	*Oak*
Plastids	+	+	+	+	+
Spores	–	+	+	+	+
Sporo-phyte dominating	–	–	+	+	+
Seeds	–	–	–	+	+
Cones	–	–	–	+	–

Literacy

1. What is the difference between cladogram and phylogenetic tree?
2. Why are reptiles closer to birds than amphibians?
3. What is an evolutionary significance of phylogenetic tree?

Research time

“Shejire” is a tree constructed by Kazakh people to show ancestors of different Kazakh families. What is the common between “shejire” and phylogenetic tree? Do a research and write an essay comparing “shejire” and phylogenetic tree. Construct or find “shejire” of your own family.

Terminology

- cladogram - кладограмма;
- compile - құрастыру / составлять;
- inclusive - қоса алғанда / включительно;
- phylogenetic tree -филогенетикалық шежіре /филогенетическое дерево;
- taxon - таксон.

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9.6 Speciation

You will:

- name the ways of speciation;
- classify the main mechanisms of speciation.

Key terms

- Species - group of organisms with similar features, which can interbreed and produce viable and fertile offspring;
- Speciation - formation of new and distinct species.

STQ

Why dogs are one species, but cheetah, jaguar and leopard are different?

Facts

However, Charles Darwin’s first work is called “The Origin of Species”, he did not explain how new species are produced. He mainly concentrated on natural selection.

Text

When you look at two organisms, it is hard to decide by comparing only their physical characteristics if they are in the same species or not. Two organisms can look similar but be in different species. For example, cheetah, leopard and jaguar look similar, but they are three different species. On the other hand, dogs look different, but they are one species.

For new species to be produced, organisms must undergo reproductive isolation. It means, they should not be able to interbreed and produce offspring. The process of formation of new species is called speciation. There are two main

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types of speciation: allopatric and sympatric.

Allopatric speciation

Two organism can be reproductively isolated by geographical barriers. For example, one part of bird population moves from the mainland to an island. It is separated from the original population. Here, on the island plants and animals are different from the mainland. These birds start to adapt to new environment and change their characteristics. Birds interbreed and over time new generations become different from the original population on the mainland. These two populations now cannot interbreed. A new species is formed.

There can be different geographical barriers, for example, large forest can become split, if part of the trees are cut. So, organisms become isolated from each other. Speciation when populations are isolated geographically is called allopatric speciation.

Sympatric speciation

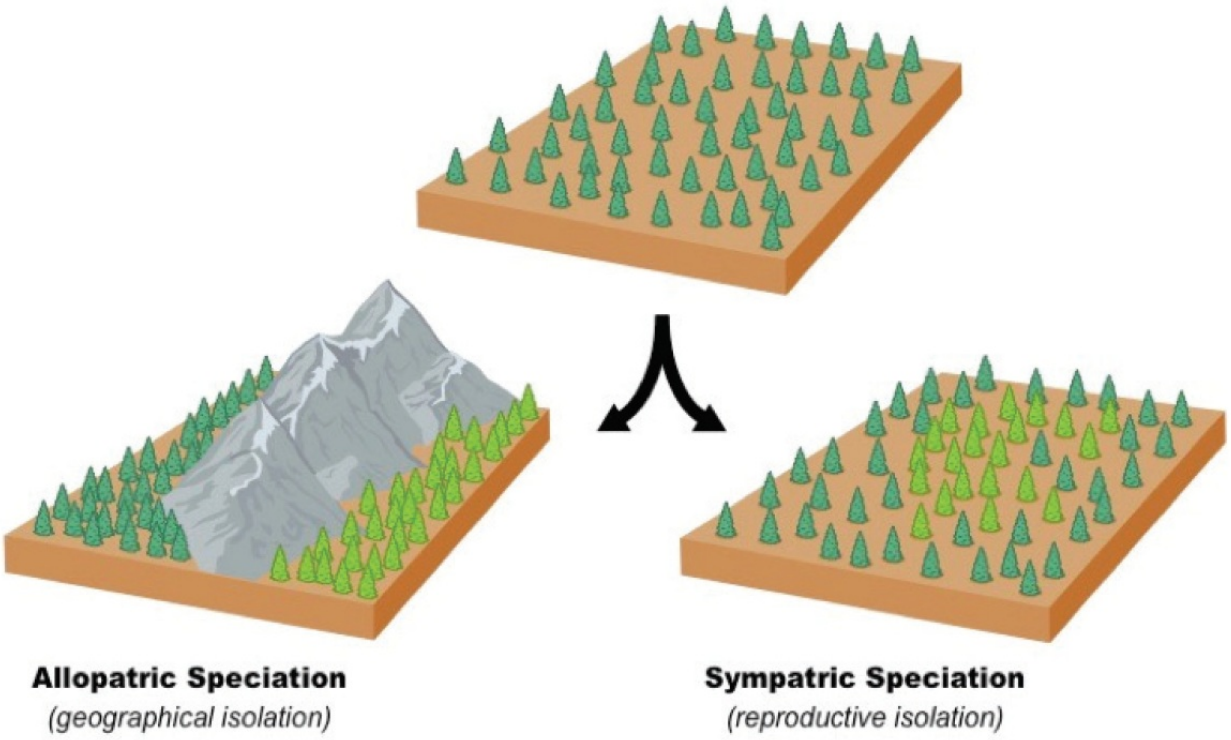
Sympatric speciation is the speciation when there is no geographical barriers. Organisms are located in one area, but become reproductively isolated. Polyploidy is the example of sympatric speciation.

Polyploid organism is an organism with more than two sets of chromosomes (3n, 4n, 5n, etc.). It can happen if chromosomes do not separate during gamete formation. This gametes will be diploid (2n). Then, when these gametes fuse, the resulting zygote will be tetraploid (4n).

This tetraploid organism can not reproduce sexually, because it is hard to undergo meiosis and produce gametes. But this organism can easily reproduce asexually and produce offspring. This offspring can grow and produce a population of tetraploid organisms. So, the resulting population cannot interbreed with the original diploid population. They are considered a new species. Polyploidy mostly happens in plants.

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Allopatric and sympatric speciation

Hybrids

Sometimes an egg of one species can be fertilized by a sperm of another species. Mostly this fertilized egg cannot develop into an organism and dies before the birth. In some cases, mating between two different species can bring to a viable offspring. The resulting organism is called a hybrid. Hybrids are mostly sterile. Mule is an example of a hybrid organism. It is a mating between a male donkey and a female mare.

Activity

Fill the table below. Compare allopatric and sympatric speciation.

Allopatric speciation	*Sympatric speciation*	
		Similarities
		Differences

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Literacy

1. What makes two species different?
2. What are main mechanisms of speciation?
3. Why hybrid organisms are sterile?

Research time

Sympatric speciation is when there is no geographical isolation between organisms. Research and find other examples of sympatric speciation except polyploidy. Describe their mechanisms and find examples.

Terminology

- interbreed - будандастыруу / скрещивать;
- mainland - құрлық / материк;
- mare - бие / кобыла;
- polyploidy - полиплоидия;
- speciation - түрдің пайда болуы / видообразование;
- sterile - ұрықсыз / стерильный;
- undergo - ұшырау / подвергаться;
- viable - өміршең / жизнеспособный.

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9.7 Selective breeding

You will:

- study ways to improve agricultural plants and animals using breeding methods.

Key terms

- Selective breeding - intentional breeding of animals or crops with particular characteristics;
- Hybrid - an offspring resulting from cross-breeding;
- Mutagenesis - a process by which the genetic information of an organism is changed, resulting in a mutation;
- Polyploid organisms - organisms containing more than two sets of chromosomes.

STQ

How to make cows to give more milk?

Facts

Pomato is a hybrid of potato and tomato. It has small tomato-like fruits, which are white inside. Pomato plant produces tomatoes on the top and potatoes underground at one time.

Text

Humans use animals and crops in their life. Firstly, people were using animals and crops raised in nature. Overtime, people understood, that it is better to grow animals and crops by themselves near the houses. People started choosing particular animals and crops to breed. This is how selective breeding started.

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Selective breeding is intentional breeding of animals or crops with particular characteristics. Due to selective breeding, alleles, which are not useful for human are eliminated from the population. Frequency of useful traits increases in the population.

For example, people need cows which give more meet. Breeders choose bigger and more muscular cows and interbreed them. Consequently, small and weak cows will die out and after several populations only big and muscular cows will be in the population.

Hybrids

Humans use other methods to produce better crop and cattle. One of them is production of hybrids, by breeding two organisms from different species. Gametes of different species fuse artificially. Resulting organism will have characteristics of both species.

The most common example of hybrid organism is a mule. It is an offspring of a male donkey and a female mare. It has characteristics of both parents. It is strong like a mare and enduring like a donkey. So, it is useful in agriculture.

Hybrids cannot reproduce sexually. So, they are mostly used in production of crops, because plants can reproduce asexually. Russian scientist Ivan Michurin cultivated different hybrids of apples, pears, cherry, and others. Also, he was one to start cultivating apricots, grapes and other southern plants in the northern climate.

Mutagenesis

Scientists artificially produce mutant organisms, which have better features than original organisms. Polyploidy is an example of production better crops via mutations. Polyploids often have more nutrients, more resistant to bad conditions and give more yield. So, scientists use mutagens, like chemicals and radiation, to stop chromosome segregation during meiosis. Resulting gametes will have more than one set of chromosomes and produce polyploid organisms after fertilization. Some varieties of apples, bananas, strawberries, watermelon, etc. are examples of polyploid plants.

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Activity

Crops and plants, which were mutated and are used as food are called genetically modified organisms (GMO). Most people think GMO are bad for human health. However, investigations showed no reports of ill effects in the human from GMO. In 2016, 120 Nobel prize winners wrote a letter to Greenpeace, the United Nations and governments around the world to stop fighting GMO.

Think about the advantages and disadvantages of GMO. Discuss it with your friends and fill the table below.

Literacy

1. What are advantages of selective breed animals and crops?
2. What are potential risks of production of mutant plants and animals?
3. What is the difference between hybrids and polyploids?

Research time

Genetic mutations are used to produce crops with beneficial characteristics. Polyploidy is one example of such a mutation. Do a research and find other examples of genetic mutations beneficial for agriculture. Which useful characteristics do they give to plants?

Terminology

- enduring - төзімді / выносливый;
- selective breeding - сұрыптау / селекция, искусственный отбор;
- yield - егін /урожай;

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9.8 Human evolution

You will:

- learn the stages of anthropogenesis.

Key terms

- Praeanthropus - the stage of human evolution which includes Australopithecus;
- Archanthropus - stage includes Homo erectus species;
- Paleoanthropus - stage includes Neanderthals;
- Neoanthrop - stage includes modern human Homo sapiens.

STQ

How did human species appear?

Facts

If we believe that all human originated in Africa, it means that human could travel through oceans 40-35 thousand years ago, because fossils of Cro-Magnon human were found in Australia and North America. Scientists believe that human used boats to reach Australia and North America.

Text

There is a theory that evolution of human began from extinct apes called Dryopithecus. They looked more like monkeys than human. They were living 25-30 million years ago in Africa and South Asia. Nowadays apes and humans originated from these organisms.

Praeanthropus

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Praeanthropus is the stage of human evolution which includes Australopithecus. They existed from 5.5 to 1.8 million years ago. From all the Praeanthropus there was one species, which was found in Olduvai Gorge in Tanzania. Near his fossils different tools from stones were found. This species was named Homo habilis. They were gathering seeds and fruits. Also, they were hunting on small animals. The structure of the skeleton of Homo habilis shows that they had bipedalism. It is believed that modern human derived from this species but other Australopithecus did not evolve.

Archanthropus

Archanthropus stage includes Homo erectus species. They were living from 1.9 to 0.5 million years ago in Europe, Asia and Africa. They were walking on two legs and used hands only for different activities. They could use fire. They learned to make better tools. Also, they learned to speak.

First Homo erectus was found by Dutch scientist Eugene Dubois in Indonesia on Java island. He named this organism Pithecanthropus erectus. After that, other fossils of Homo erectus were found in Germany and China. The “chinese” organism was called Sinanthropus pekinensis.

Paleoanthrpous

The Paleoanthropus stage includes Neanderthals. They were living from 150 to 35 thousand years ago. First they were found in Germany near Dusseldorf. After that their fossils were found in different parts of Europe, Asia and Africa.

Neanderthals were very strong. Their bones had dense structure. They were living in caves and used animal skins to cover themselves. They were hunting on bigger animals like mammoths. They had better tools from stones and bones. They used fire to cook and make their houses warmer.

Also, in caves of Neanderthals some ritual burials were found. It means that they had some religions. It is believed that they were worshipping different animals like cave bears.

Neoanthrop

Neoanthrop stage includes modern human Homo sapiens. They were living 40-

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35 thousand years ago. First fossil of Homo sapiens were found in France in Cro-Magnon site. They were named by the place they were found in - Cro-Magnon. Fossil of Cro-Magnon were found in Europe, Asia, Africa, Australia and North America.

Physically Cro-Magnon looked like modern human. They were good hunters. It is believed that they exterminated large animals like Mammoths. Because there were less mammoths, Cro-Magnon started domesticating different animals. Also, they started growing plants.

Activity

Fill the table below.

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	Praeanthropus	Archanthropus	Paleoanthrpous	Neoanthrop
Neanderthals				
Homo sapiens				
Homo habilis				
Australopithecus				
Homo erectus				
Cro-Magnon				
Lived in Asia				
Lived in Europe				
Lived in Africa				
Lived in Australia				
Lived in America				
Domestication of animals				
Hunting on big animals				
Gathering fruits				
Agriculture				
Used tools				

Literacy

1. Where first human originated?
2. In which areas did ancient human live? What are advantages of these areas?
3. What were the first human tools made of?

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Research time

Construct a phylogenetic tree of genus Homo.

Terminology

- ape - маймыл тектес адам / человекообразная обезьяна;
- bipedalism - тік жүру / прямохождение;
- burial - жерлеу / захоронение;
- exterminate - жою / истреблять;
- worship - табыну / поклоняться.

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Problems

Test questions with one right answer

1. What does each branch on the phylogenetic tree represent?

A) a common ancestor

B) a derived characteristic

C) a shared characteristic

D) a taxon

E) LUCA

2. Intentional breeding of animals or crops with particular characteristics?

A) natural selection

B) bottleneck effect

C) selective breeding

D) mutagenesis

E) polyploidy

3. Stage includes Neanderthals?

A) Praeanthropus

B) Archanthropus

C) Paleoanthropus

D) Neoanthrop

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E) Australopithecus

4. When did first eukaryotic organisms occur?

A) 3.5 billion years ago

B) 2.7 billion years ago

C) 1.8 billion years ago

D) 1.2 billion years ago

E) 600 million years ago

Test questions with several (max 3) right answers

1. Causes of genetic variation

A) mitosis

B) DNA replication

C) crossing-over

D) DNA repair

E) development

F) independent assortment

G) fertilization

H) asexual reproduction

2. Find the examples of homology:

A) whale fin and human arm

B) butterfly wing and hawk wing

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C) dolphin fin and a penguin fin

D) potato and a sweet potato

E) platypus egg and duck egg

F) human tailbone and monkey tail

G) human leg and ant leg

H) leaves of a Venus fly trap and cactus needles

3. Which of the following decreases genetic variation?

A) natural selection

B) independent assortment

C) fertilization

D) meiosis

E) mutation

F) bottleneck effect

G) reproduction

H) founder effect

Matching questions

1. Match the terms with their definitions:

1. Natural selection

2. Bottleneck effect

3. Founder effect

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A) a change in DNA of an organism

B) A sudden catastrophe, such as fire or flood.

C) shuffling existing alleles to produce new combinations of genes

D) increases the genetic variation

E) the process, when more adapted individuals survive and reproduce

F) When a few individuals become isolated from a population, they start reproducing with each other forming a new population

2. Match the terms with their definitions:

1. Allopatric speciation

2. Sympatric speciation

3. Hybridization

A) mating between two different species

B) there is no geographical barrier

C) organisms containing more than two sets of chromosomes

D) Two organism can be reproductively isolated by geographical barriers

E) competition between organisms of a population

F) change in the frequency of an existing allele in a population due to random sampling of organisms.

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